We can even quantify what “hot and dense” means, and trace the history of the universe backward from the cool and pleasant cosmos we are enjoying now to a pressure-cooker inferno so extreme it shatters our understanding of the laws of physics.

This isn’t just a theoretical exercise, though. It’s one thing to mathematically extrapolate expansion and derive higher pressures and temperatures; it’s another to see this infernoverse[fn7] directly.

THE COSMIC MICROWAVE BACKGROUND

The story of how we went from thinking about the Big Bang to seeing it is a classic tale of serendipitous discovery in cosmology. In 1965, a physicist named Jim Peebles at Princeton University was doing the calculations, dialing back the cosmic expansion, and coming to the startling conclusion that radiation from the Big Bang should still be streaming through the universe today. Moreover, it should be detectable. He calculated the expected frequency and intensity of that radiation and teamed up with colleagues Robert Dicke and David Wilkinson to start building an instrument to measure it. Meanwhile, unbeknownst to them, just down the road at Bell Labs, a couple of astronomers named Arno Penzias and Robert Wilson were gearing up to do some astronomy with a microwave detector that had previously been used for commercial purposes. (Microwaves are just a kind of light on the electromagnetic spectrum, higher frequency than radio but lower frequency than infrared or visible light.) When Penzias and Wilson, totally uninterested in commercial applications and keen on studying the sky, were calibrating the instrument for their research, they found a weird humming in the feed. Apparently it hadn’t interfered with the previous use of the telescope, detecting communication signals bounced off high-atmosphere balloons, so those users had ignored it. But this was for *science*, and it had to be fixed. The hum appeared no matter which direction they pointed the detector and was, by all accounts, extremely inconvenient.

Telescope interference is a common problem during the calibration phase of an observing run, and there are a lot of ways it can happen. There could be a loose cable somewhere, or radio interference coming from some transmitter nearby, or any number of little mechanical annoyances. (A recent big break in radio astronomy involved the discovery that tantalizing bursts of radiation seen by the Parkes radio telescope were actually due to